

Answer all the questions from each group

Group A

Q1 Answer any FIVE

5x2=10

- What is the purpose of the Program Counter (PC)?
- Give an example of an instruction which uses immediate addressing mode in any contemporary instruction set architecture.
- Describe what is IEEE 754.
- What is the hit rate of a cache?
- What unit of measurement is used for the clock period of a processor?
- Given a 4-stage pipeline with stage delays of 150, 120, 160, and 140 nanoseconds, and a 5-nanosecond delay for each pipeline register, calculate the total time required to execute 100 instructions assuming no instruction dependencies.
- What is the full form of the abbreviation SISD?
- If a cache is n -way set associative and contains m sets, what is the value of n when the cache is direct-mapped and when it is fully associative?

Q2 Answer any FOUR

4x5=20

- Draw the components of a computer using the Von Neumann Architecture and how they are connected using a diagram.
- Describe at least three commonly used addressing modes in contemporary ISAs.
- Calculate $(+46)_2 + (-42)_2$ using 2's complement notation.
- Derive the speedup formula of a computer system using Amdahl's Law. Assume we aim to improve a processor used in web serving. The upgraded processor is ten times faster in computation compared to the original one. If the original processor spends 40% of the time on computation and 60% on I/O, calculate the overall speedup after the enhancement.
- Consider a processor P1 that is running a program with 2500 instructions. P1 has a 4 GHz clock rate and an IPC of 0.8. How many seconds will the CPU take to complete the execution of the program?
- Describe the classification in Flynn's taxonomy.

Q3 Answer any TWO

2x10=20

- i) Define the write-allocate and no-write-allocate cache policies. Consider a fully associative write-back cache with many entries, initially empty. Given the following sequence of five memory operations (addresses in square brackets):

```
Write Mem[100];
Write Mem[100];
Read Mem[200];
Write Mem[200];
Write Mem[100];
```

Determine the number of cache hits and misses for both no-write-allocate and write-allocate strategies.

- ii) Between a 16 KB instruction cache paired with a 16 KB data cache and a unified 32 KB cache, which configuration results in a lower miss rate? Use the miss rates provided in the given figure 1, assuming that 36% of instructions are data transfers. A cache hit takes 1 clock cycle, and a miss incurs a penalty of 100 clock cycles. In a unified cache with a single port, a load or store hit adds 1 extra clock cycle when two simultaneous accesses occur. Compute the average memory access time for each configuration. Assume write-through caches with a write buffer, and ignore write buffer stalls.

Size (KB)	Instruction cache	Data cache	Unified cache
16	3.82	40.9	51.0
32	1.36	38.4	43.3

Figure 1: Miss per 1000 instructions for instruction, data, and unified caches of different sizes. The percentage of instruction references is about 74%. The data are for two-way associative caches with 64-byte blocks.

- Convert the following 32-bit floating-point number to decimal notation:

sign	exponent	mantissa
0	10000110	110101000000000000000000
1 bit	8 bit	23 bit

32 bit

Group B

Q4 Answer any **FIVE**

5x2=10

- a) Describe the purpose of an ALU.
- b) What is an implementation of an ISA?
- c) How do stack overflows occur?
- d) Give one example of a scheme that may be used to determine which block needs to be replaced in fully associative cache mapping
- e) What are pipeline stages?
- f) Describe what is superscalar processing.
- g) What are interrupts?
- h) Given 4KB pages, how many bytes can a 16-way set-associative VIPT (Virtually Indexed, Physically Tagged) cache hold?

Q5 Answer any **FOUR**

4x5=20

- a) Differentiate between address, data and control buses.
- b) What types of information are contained in stack memory?
- c) Consider a program where 80% of the work can be parallelized by using parallel processing. How many processing units would be needed to get a speedup of four times and what it would be for a single processing unit?
- d) Differentiate between write-back and write-through cache writing schemes.
- e) Describe how operand forwarding works.
- f) Why does vector processing increase the performance of a processor?

Q6 Answer any **TWO**

2x10=20

- a) For a directly mapped cache design with a 64-bit address, the following bits of the address are used to access the cache.

Tag	Index	Offset
63-10	9-5	4-0

Beginning from power on, the following byte-addressed cache references were recorded.

Address												
Hex	00	04	10	84	E8	A0	400	1E	8C	C1C	B4	884
Dec	0	4	16	132	232	160	1024	30	140	3100	180	2180

For each reference, list/tabulate the following:

3 x 3 = 9

- i) Its tag, index and offset
- ii) Whether it was a hit (aka data was in the cache) or cache miss (aka no data was in the cache prior)
- iii) Which bytes were replaced (if any)

What is the hit ratio?

- b) Consider a computer running the following instructions (Note: add and addi both add the second and third operands and save the result in the first operand). Assume that x11 is initialized to 11 and x12 is initialized to 22. Suppose you executed the instructions listed here in a five-stage pipeline containing the pipeline stages IF (instruction fetch), ID (instruction decode), EX (execute), MEM (memory access), WB (write back).

```
addi x11, x12, 5
add x13, x11, x12
addi x14, x11, 15
add x15, x11, x11
```

First find where the data hazards are occurring here. Then solve the data hazard by adding NOP operations to stall the pipeline. Explain how your solution solves the data hazard.

- c) A CPU generates 48-bit virtual addresses, and the physical memory is 4 GB in size.
 - i) If the page size is 64 KB, how many bits are used for the page offset and the physical page number? Also, calculate the total number of entries in the page table.
 - ii) What are the advantages of using a Translation Lookaside Buffer (TLB)? Describe how a TLB works using a diagram.
 - iii) What is a multi-level page table, and why is it necessary in modern systems?